

Gene Regulation

You have already learned the flow of information from gene to protein, from genotype to phenotype. Every living thing has molecular mechanisms that open and shut this pathway. This regulation is one of the very basic mechanisms that living things use to respond to changes in their environment and regulate their own internal machinery through negative and positive feedback.

In theory, we could regulate this process at almost any step. We could change DNA sequences, alter the genetic information itself - for instance, to change the binding of the transcription machinery at promoters. We could modify the stability of the mRNAs produced. We could modulate the splicing, or export of transcripts from the nucleus, or alter the binding efficiency of the ribosomes on their mRNA templates. Indeed, if you can imagine a possible molecular mechanism to influence the flow of information from genotype to phenotype then there is probably at least one organism or process that uses it. However, regulating the initiation of transcription is by far the most prominent. We are going to focus on this process, but do not begin to believe that, because of this focus, other mechanisms are not at work.

Unlike the other topics covered so far: replication, transcription, and translation - gene regulation in prokaryotes and eukaryotes is worth considering separately from the outset. I want you to follow a particular study order. Focus on the Prokaryotic Lac operon before you start considering Eukaryotes. And, because of the deep divergence between the two, remember to be aware at all times as you study whether the material applies to Prokaryotes or Eukaryotes. So start with AN OVERVIEW OF GENE CONTROL, then move to the first part of TRANSCRIPTION REGULATORS SWITCH GENES ON AND OFF in **MBoC** and **Cooper** Lac operon (posted on Website). Nail operon control, then loop back to CONTROL OF TRANSCRIPTION BY SEQUENCE-SPECIFIC DNA-BINDING PROTEINS in **MBoC**. Surprising to me, the USMLE can hit you with some of the core DNA-binding domain names in Panel 7-1. While they may do that, they will likely blow over an incredibly important point in this section that you need to absorb - proteins can recognize DNA sequence by 'reading' the coded surface present in the major and minor DNA grooves. This is important for a lot more than just regulating gene expression. Now move forward to consider the condensation of the transcription machinery on Eukaryotic promoters. Finally, dig into the next level of gene regulation that occurs in eukaryotes only, the relation between chromatic structure/modification and the ability of the DNA therein to be transcribed. I will use its relevance to gene expression to push you to look at the higher order structures of DNA in the Eukaryotic nucleus.

Reading:

MBoC(6th) Ch7: Control of Gene Expression, **Ch4:** Nucleosomes are a basic unit of Eukaryotic chromosome structure and then CHROMATIN STRUCTURE AND FUNCTION (See annotated TOC on Moodle). **Cooper (3rd).** Lac Operon - Posted to website.

There is more material covered in the chapters than you need to know. Use the text and illustrations below to help focus.

Need to know and understand

Genes are often regulated in response to external signal. It may be a change in environment, eg. presence of lactose, or the need to modify a differentiation phenotype, e.g. during embryonic development. The pathway from environment to gene expression is an example of Signal Transduction.

Prokaryotic Operon regulation

- Cistron, polycistronic
- Promoter
- Operator
- Inhibitor/Repressor
- Inducer
- Lac operon (Lac I, CRP site, O_{lac} , P_{lac} , Lac-Z, -Y, -A)
- Catabolite repression
- catabolite gene activator protein (CAP)
- cAMP

Eukaryotic gene regulation.

- assembling the pol II initiation complex at the promoter

On the DNA

- Two main types of promoters - constitutive and regulated
- core, proximal, and distal (enhancers) promoter elements
- Core: TATA box, CAAT box, Initiator region (INR), Downstream Promoter Element (DPE)
- Proximal: GC box
- Distal: enhancers
- Transcription factor binding sites, upstream activating sequences (UAS)
- consensus sequence

Proteins that bind to DNA

- Transcription factors (note that this term is so broad as to be almost non-descriptive. Almost anything that affects the initiation of transcription has been called a transcription factor, although they may function in very different ways.)

- Many transcription factors contain protein domains that bind to specific DNA sequences. Most contain members of a few families of DNA binding domains: homeodomain, zinc-finger, beta-sheet DNA-binding, leucine zipper, helix loop helix (HLH), Helix-Turn-Helix (HTH). They usually contain another domain that mediates protein-protein interaction.

- General transcription factors are part of the PolIII holoenzyme and bind to core-promoter elements

- activators/repressors/co-activators/co-repressors

- enhancesome. Complex of small stretch of DNA containing multiple transcription factor binding sites and its protein residents.

Gene regulation by alternative splicing

Gene regulation by alternative promoters and alternative termination

Epigenetics

- Modulation of chromatin structure (if you can't get to the promoter, you can't transcribe the gene).

- Histone deacetylation decreases transcription

- Histone acetylation increases transcription

- X-inactivation

heterochromatinization

Methylation: Mostly the C of CG dinucleotides

Pattern of methylation changes during embryonic development

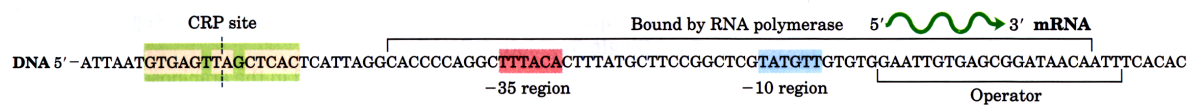
Sperm and Oocyte DNA are differentially methylated

Linkage between DNA methylation and Histone acetylation

Post-transcriptional regulation

Gene Regulation in Medicine. Mutations in transcription factors often lead to developmental abnormalities. Epigenetic modifications are starting to be appreciated in a variety of complex diseases, such as diabetes and cancer.

Focus Illustrations: Prokaryotes



The lac operon promoter region showing the actual sequences involved in regulating operon expression. (As is usual in these kinds of figures, they are showing only one strand of the DNA and, as usual again, the coding-strand with the 5' end to the left. Remember, it's actually double-stranded DNA.)

Focus illustrations: Source: The Cell: 3rd Edition (2004) Cooper. Sinauer, Sunderland, MA

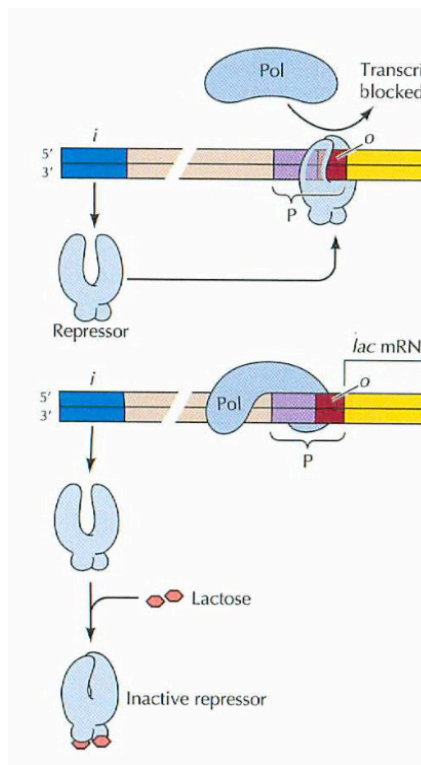


Figure 6.9 Negative control of the lac operon. The *i* gene encodes a repressor which, in the absence of lactose (top), binds to the operator (*o*) and interferes with the binding of RNA polymerase to the promoter, blocking transcription of the three structural genes (beta galactosidase(*z*), lac permease (*y*), transacetylase(*a*)). Lactose induces expression of the operon by binding to the repressor (bottom), which prevents the repressor from binding to the operator. P=promoter; Pol=RNA polymerase.

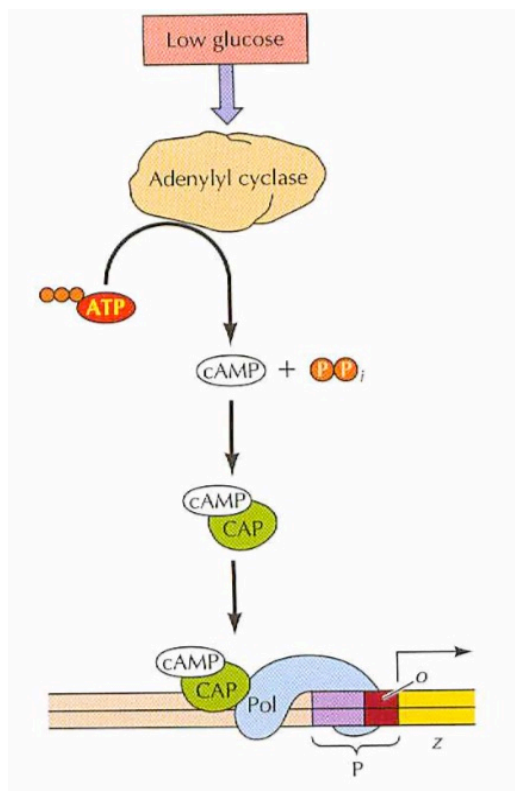


Figure 6.10 Positive control of the Lac operon by glucose. Low levels of glucose activate adenylate cyclase, which converts ATP to cyclic AMP (cAMP). cAMP then binds to the catabolite activator protein (CAP) and stimulates its binding to regulatory sequences (CRP site on the previous page) of various operons concerned with the metabolism of alternative sugars, lactose being one such. CAP sticks to the alpha subunits of RNA polymerase, thus assisting it in sticking to the promoter. The polymerase needs this because P_{lac} is a poor promoter; the interaction between it and the sigma subunit of RNA polymerase is very weak.

Focus illustrations: Eukaryotes Source: Molecular Biology of the Cell: Sixth Edition
(2015) Alberts et al., Garland Science, NY

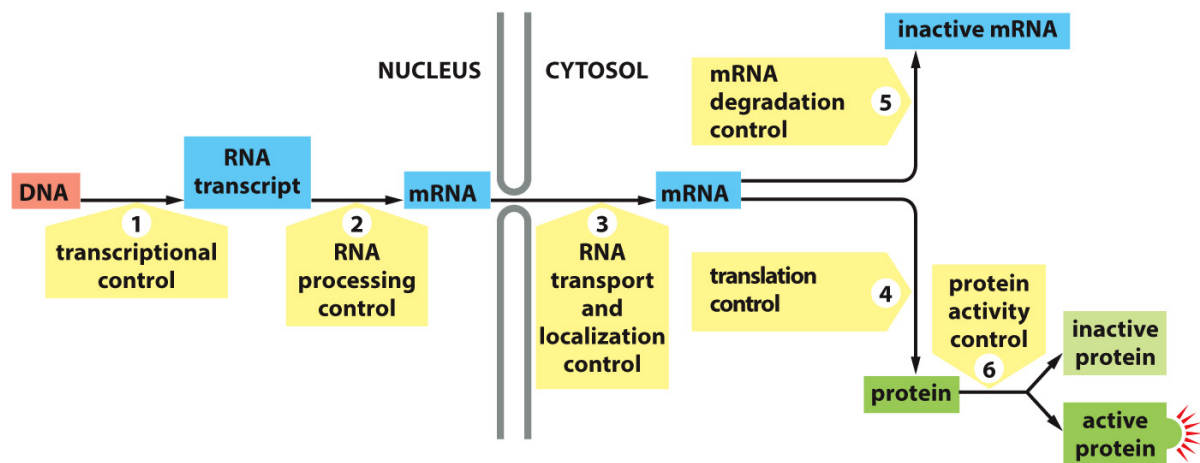


Figure 7-5 Molecular Biology of the Cell 6e (© Garland Science 2015)

Our focus on regulation of transcription excludes many other possibilities. Figure 7–5 Six steps at which eukaryotic gene expression can be controlled. Controls that operate at steps 1 through 5 are discussed in this chapter. Step 6, the regulation of protein activity, occurs largely through covalent post-translational modifications including phosphorylation, acetylation, and ubiquitylation (see table 3–3, p. 165). Step 6 was introduced in Chapter 3 and is subsequently discussed in many chapters throughout the book.

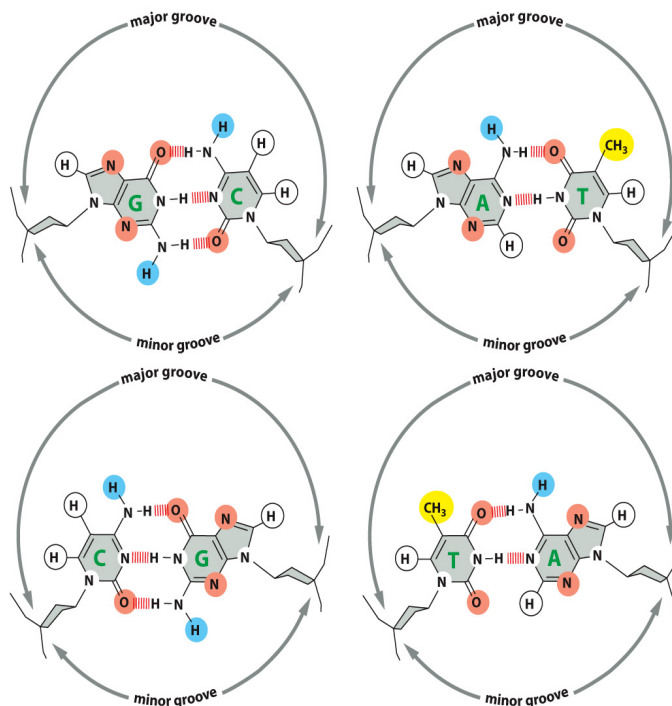


Figure 7-7 Molecular Biology of the Cell 6e (© Garland Science 2015)

Understand the concept here, not the details!

Figure 7–7 How the different base pairs in DNA can be recognized from their edges without the need to open the double helix. the four possible configurations of base pairs are shown, with potential hydrogen-bond donors indicated in blue, potential hydrogen-bond acceptors in red, and hydrogen bonds of the base pairs themselves as a series of short, parallel red lines. Methyl groups, which form hydrophobic protuberances, are shown in yellow, and hydrogen atoms that are attached to carbons, and are therefore unavailable for hydrogen-bonding, are white. from the major groove, each of the four base-pair configurations projects a unique pattern of features. (from C. Branden and J. tooze, introduction to protein

Structure, 2nd ed. new York: Garland publishing, 1999.)

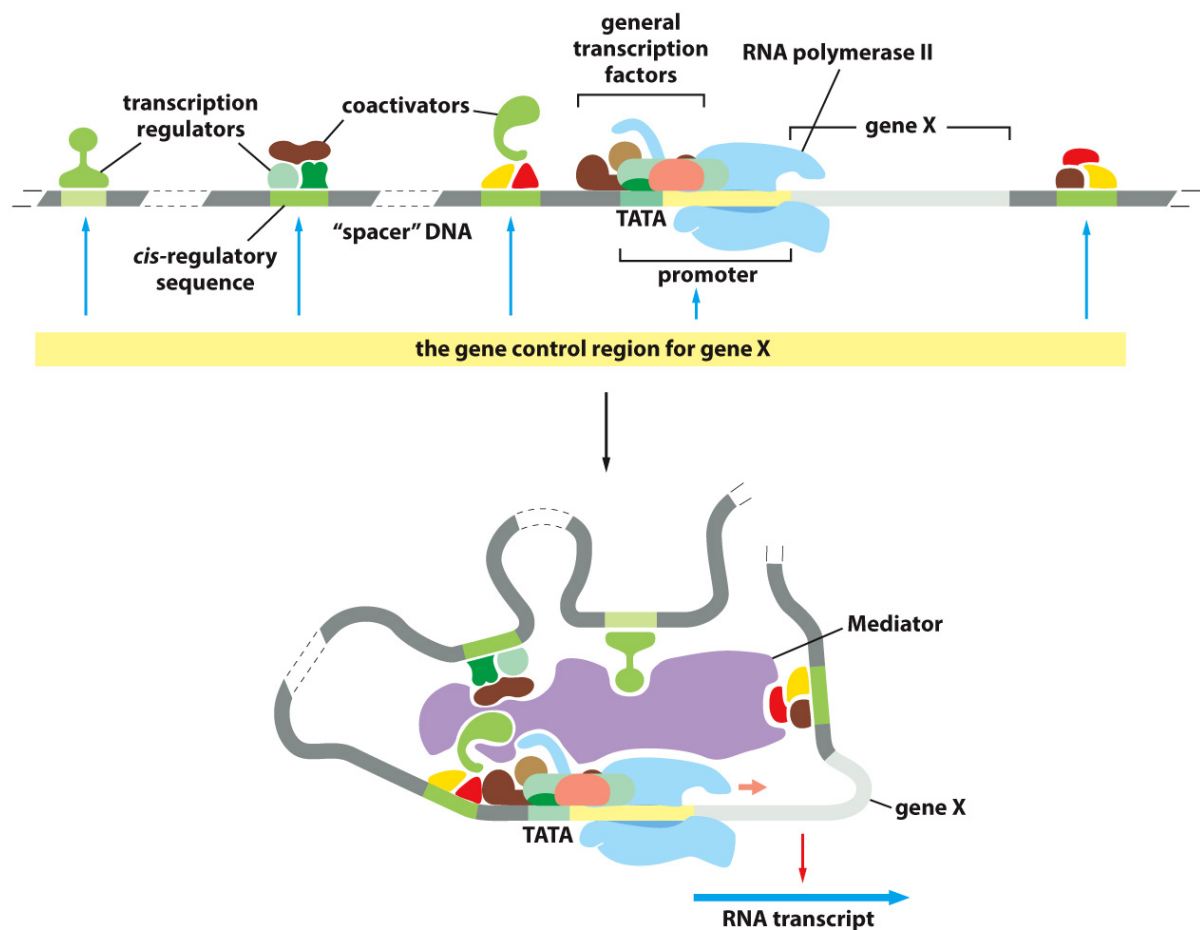


Figure 7-17 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 7–17 The gene control region for a typical eukaryotic gene. the promoter is the DNA sequence where the general transcription factors and the polymerase assemble (see figure 6–15). The cis-regulatory sequences are binding sites for transcription regulators, whose presence on the DNA affects the rate of transcription initiation. These sequences can be located adjacent to the promoter, far upstream of it, or even within introns or entirely downstream of the gene. the broken stretches of DNA signify that the length of DNA between the cis-regulatory sequences and the start of transcription varies, sometimes reaching tens of thousands of nucleotide pairs in length. The TATA box is a DNA recognition sequence for the general transcription factor TFIID. As shown in the lower panel, DNA looping allows transcription regulators bound at any of these positions to interact with the proteins that assemble at the promoter. Many transcription regulators act through Mediator (described in Chapter 6), while some interact with the general transcription factors and RNA polymerase directly. Transcription regulators also act by recruiting proteins that alter the chromatin structure of the promoter (not shown, but discussed below). Whereas Mediator and the general transcription factors are the same for all RNA polymerase II-transcribed genes, the transcription regulators and the locations of their binding sites relative to the promoter differ for each gene.

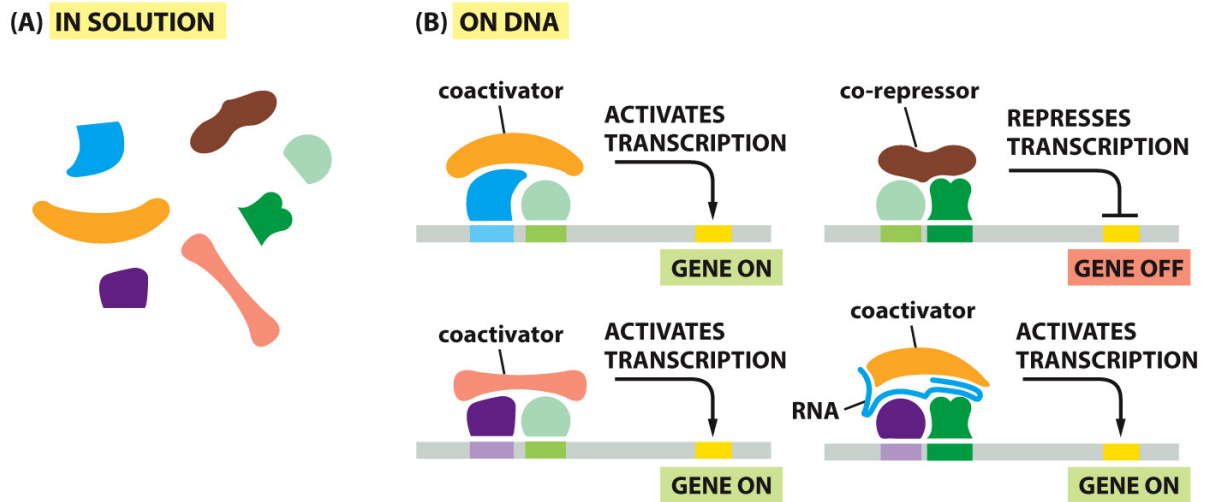


Figure 7-18 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 7–18 Eukaryotic transcription regulators assemble into complexes on DNA. (A) Seven transcription regulators are shown. The nature and function of the complex they form depend on the specific cis-regulatory sequences that seed their assembly. (B) Some assembled complexes activate gene transcription, while another represses transcription. Note that the light green and dark green proteins are shared by both activating and repressing complexes. Proteins that do not themselves bind DNA but assemble on other DNA-bound transcription regulators are termed co-activators or co-repressors. In some cases (lower right), RNA molecules are found in these assemblies. As described later in this chapter, these RNAs often act as scaffolds to hold a group of proteins together.

Focus illustrations: Eukaryotes, Epigenetics: Source: Molecular Biology of the Cell: Sixth Edition (2015) Alberts et al., Garland Science, NY

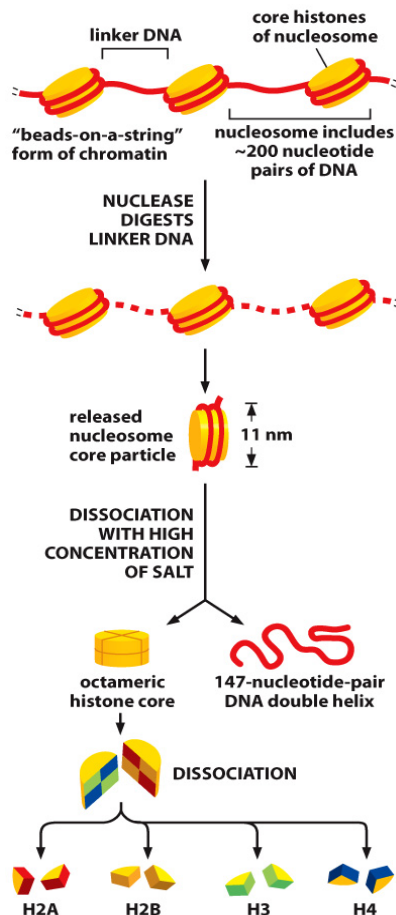


Figure 4-22 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 4–22 Structural organization of the nucleosome. A nucleosome contains a protein core made of eight histone molecules. In biochemical experiments, the nucleosome core particle can be released from isolated chromatin by digestion of the linker DNA with a nuclease, an enzyme that breaks down DNA. (The nuclease can degrade the exposed linker DNA but cannot attack the DNA wound tightly around the nucleosome core.) After dissociation of the isolated nucleosome into its protein core and DNA, the length of the DNA that was wound around the core can be determined. This length of 147 nucleotide pairs is sufficient to wrap 1.7 times around the histone core.

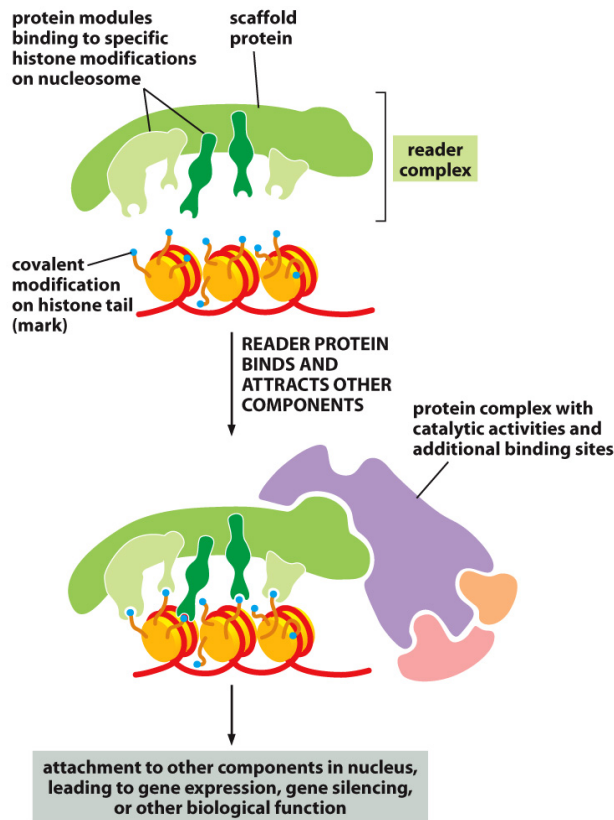


Figure 4-38 Molecular Biology of the Cell 6e (© Garland Science 2015)

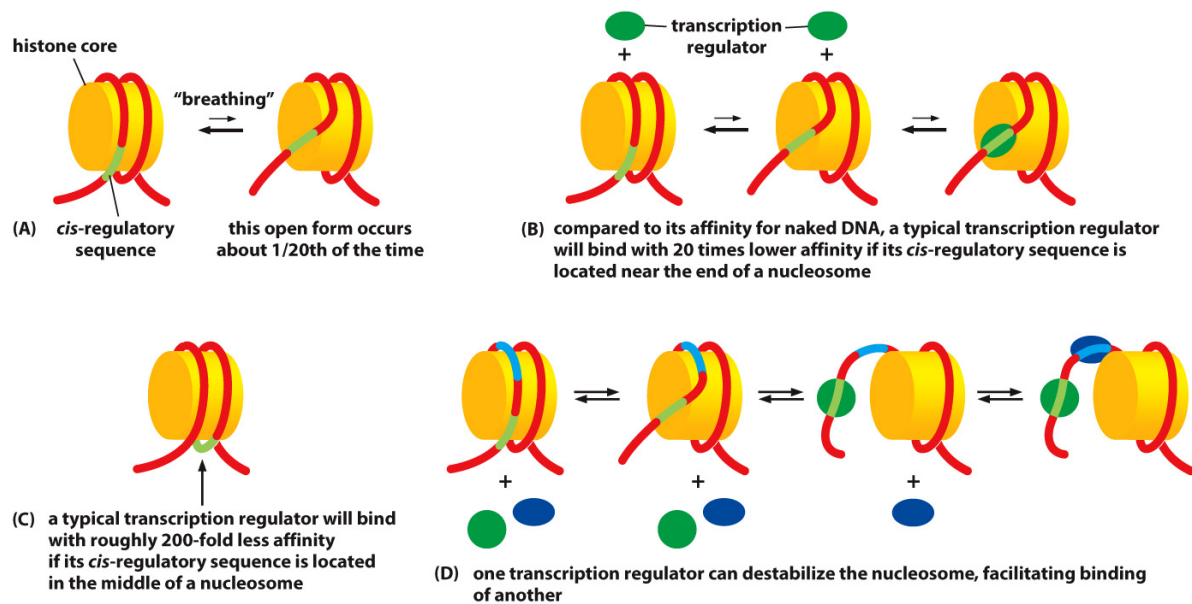


Figure 7-11 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 7-11 How nucleosomes effect the binding of transcription regulators.

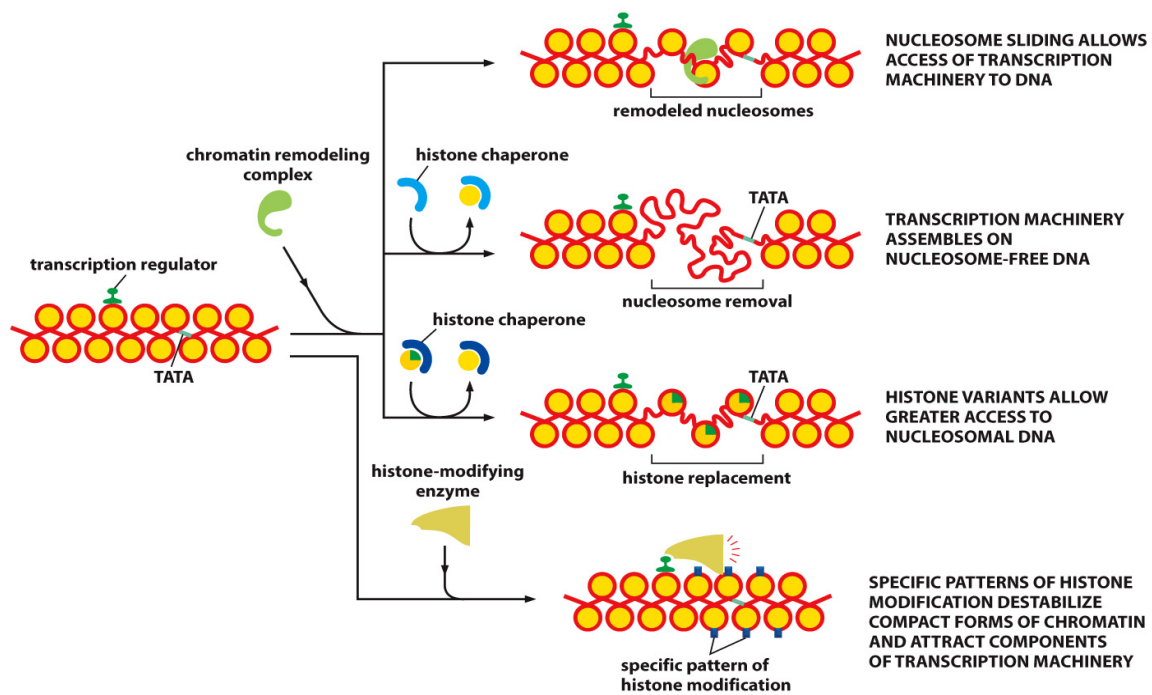


Figure 7-19 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 7–19 Eukaryotic transcription activator proteins direct local alterations in chromatin structure. Nucleosome remodeling, nucleosome removal, histone replacement, and certain types of histone modifications favor transcription initiation (see figure 4–39). these alterations increase the accessibility of DNA and facilitate the binding of RNA polymerase and the general transcription factors.

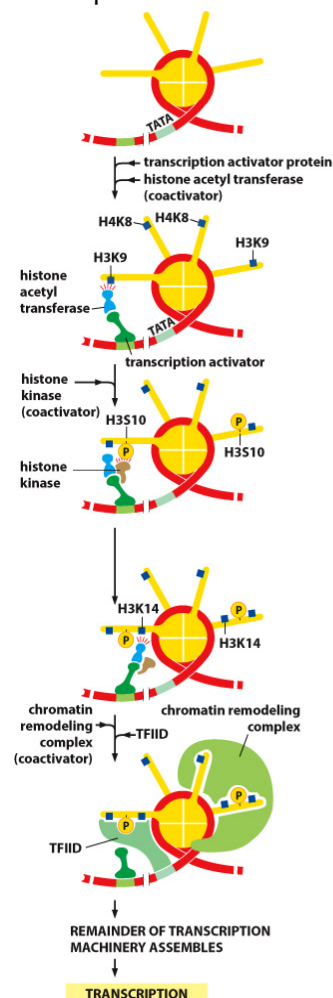


Figure 7-20 Molecular Biology of the Cell 6e (© Garland Science 2015)

Get the idea here, not the details!

Figure 7–20 Successive histone modifications during transcription initiation. in this example, taken from the human interferon gene promoter, a transcription activator binds to DNA packaged into chromatin and attracts a histone acetyl transferase that acetylates lysine 9 of histone h3 and lysine 8 of histone h4. then a histone kinase, also attracted by the transcription activator, phosphorylates serine 10 of histone h3 but it can only do so after lysine 9 has been acetylated. this serine modification signals the histone acetyl transferase to acetylate position K14 of histone h3. next, the general transcription factor tfIID and a chromatin remodeling complex bind to the chromatin to promote the subsequent steps of transcription initiation. tfIID and the remodeling complex both recognize acetylated histone tails through a bromodomain, a protein domain specialized to read this particular mark on histones; a bromodomain is carried in a subunit of each protein complex. The histone acetyl transferase, the histone kinase, and the chromatin remodeling complex are all coactivators. the order of events shown applies to a specific promoter; at other genes, the steps may occur in a different order or individual steps may be omitted altogether. (Adapted from t. Agalioti, G. Chen and D. thanos, Cell 111:381–392, 2002. with permission from elsevier.)

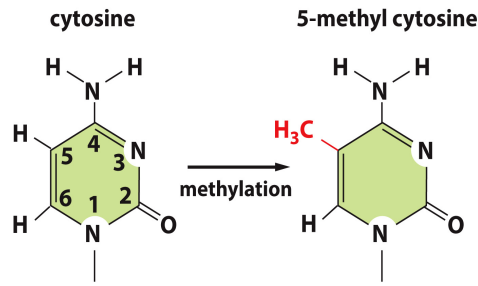


Figure 7-43 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 7–43 Formation of 5-methyl cytosine occurs by methylation of a cytosine base in the DNA double helix. In vertebrates, this event is largely confined to selected cytosine (C) nucleotides located in the sequence CG. CG sequences are sometimes denoted as CpG sequences, where the p indicates a phosphate linkage to distinguish it from a CG base pair.

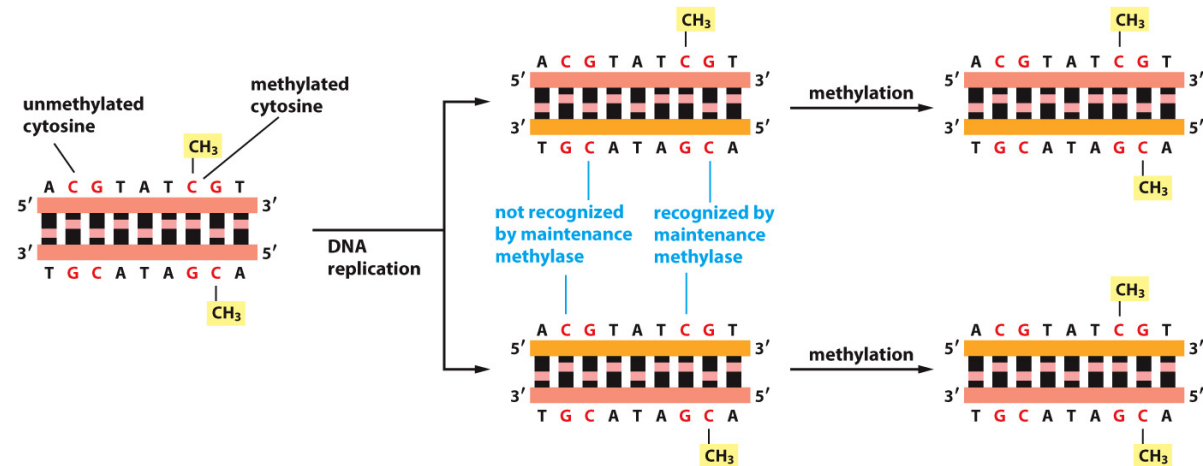


Figure 7-44 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 7–44 How DNA methylation patterns are faithfully inherited. In vertebrate DNA, a large fraction of the cytosine nucleotides in the sequence CG is methylated. Because of the existence of a methyl-directed methylating enzyme (the maintenance methyl transferase), once a pattern of DNA methylation is established, that pattern of methylation is inherited in the progeny DNA, as shown.

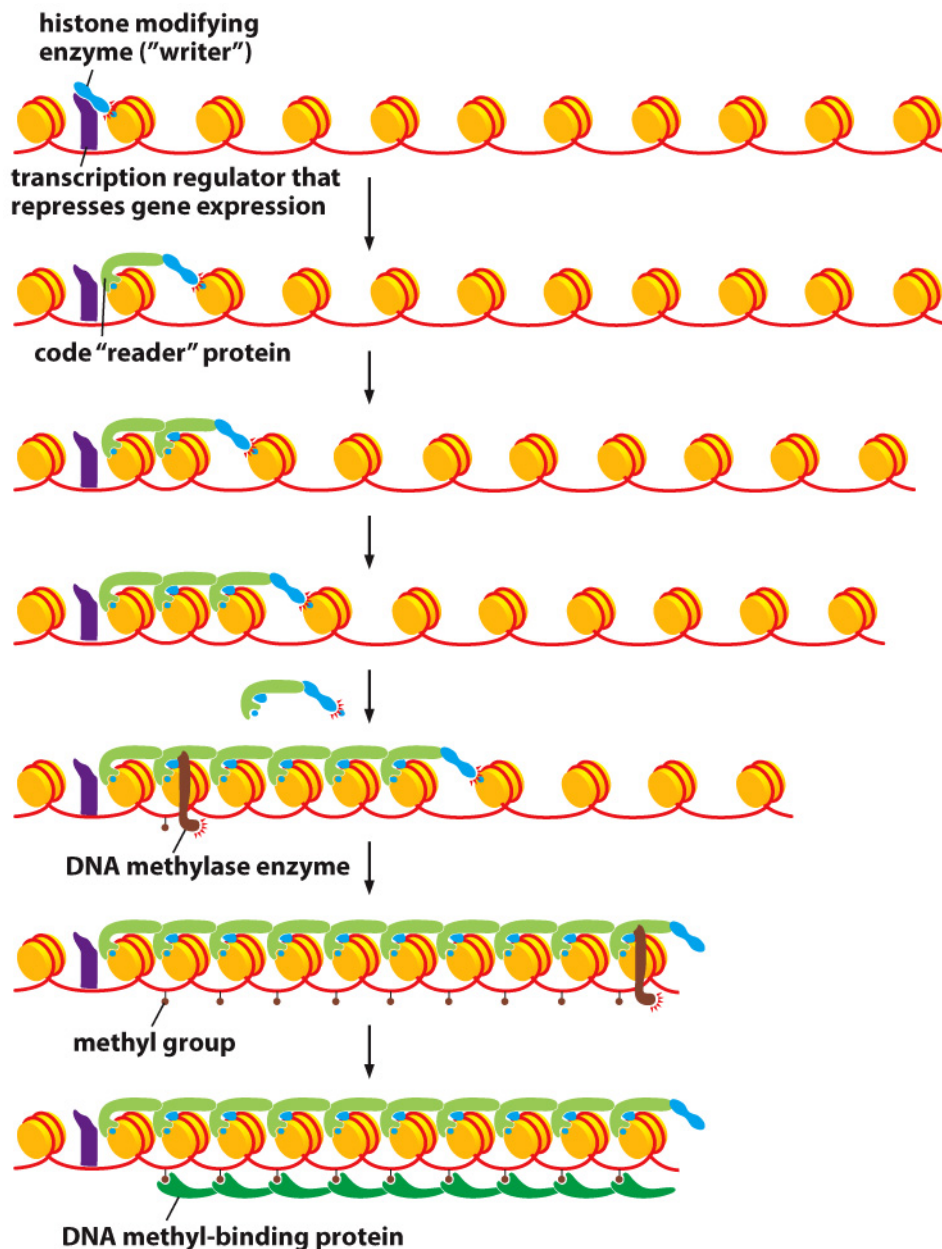


Figure 7-45 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 7–45 Multiple mechanisms contribute to stable gene repression. In this schematic example, histone reader and writer proteins (discussed in Chapter 4), under the direction of transcription regulators, establish a repressive form of chromatin. A *de novo* DNA methylase is attracted by the histone reader and methylates nearby cytosines in DNA, which are, in turn, bound by DNA methyl-binding proteins. During DNA replication, some of the modified (blue dot) histones will be inherited by one daughter chromosome, some by the other, and in each daughter they can induce reconstruction of the same pattern of chromatin modifications (discussed in Chapter 4). At the same time, the mechanism shown in figure 7–44 will cause both daughter chromosomes to inherit the same methylation pattern. In these cases where DNA methylation stimulates the activity of the histone writer, the two inheritance mechanisms will be mutually reinforcing. This scheme can account for the inheritance by daughter cells of both the histone and the DNA modifications. It can also explain the tendency of some chromatin modifications to spread along a chromosome (see figure 4–44).